

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/odl20lite, x86-64 Linux, GCC 15

Verdict

This is no longer an archive question. In a day the tree went from an unvalidated 2017 propagator to a reproducible **5.7 cm out-of-sample orbit determination result** against IGS final orbits, from a single untuned configuration, with the spacecraft's area-to-mass ratio recovered consistently to 0.03% across days. The ray-traced radiation pressure assets are conclusively gone — and the surprise is that they mostly do not matter: one estimated parameter closes most of the gap they existed to close. The asset is the demonstrated trustworthiness, not the missing physics.

What changed since revision 2

Revision 2 recommended estimating radiation pressure as a parameter rather than hard-coding an area, and held the ray-tracer pending that result. The result is in, and it is decisive on both counts.

arc	6 param, tuned	7 param, untuned	factor
GPS G01, 2023-01-22	0.343 m	0.0645 m	5.3×
GPS G01, 2023-01-23 (<i>out-of-sample</i>)	0.334 m	0.0573 m	5.8×
GPS G05, 2023-01-22	3.200 m	0.1870 m	17×

The left column required the spacecraft area to be hand-tuned per satellite. The right column is one generic, deliberately untuned configuration — 10 m² and 1000 kg — serving every arc, converging in three iterations from a cold start on all of them.

For scale: IGS final orbits, used here as truth, are themselves quoted at about 2.5 cm. A 5.7 cm fit residual is therefore within roughly a factor of two of the reference product's own accuracy. This is not better than what analysis centres achieve — they reach a few centimetres with far more elaborate parameterisation. It is close enough to reference practice to be credible, obtained from seven parameters, in a tree that did not build at all a few weeks ago.

The two results that matter more than the RMS

The estimated parameter is a property of the spacecraft, not of the arc. The same satellite fitted on two different days returns scale factors agreeing to **0.03%**. Nothing was tuned to produce that agreement.

The parameter located a discrepancy rather than absorbing it. G05 returns a different scale from G01 — implied A/m of 0.0190 against 0.0206 m²/kg, both near the published GPS figure of ~0.020 — and that difference is precisely why the six-state fit was ten times worse on G05. A parameter that had quietly swallowed the discrepancy would have been worth much less than one that exposed it.

Together these are the difference between a curve fit and a measurement. They are also the reason the accuracy claim is defensible to someone hostile: the model recovers a physical property of the satellite, independently, on arcs it was not tuned against.

The ray-tracer: decided against

Revision 2 held this open pending the number above. The number has arrived and it closes the question.

A generic cannonball with one estimated parameter reaches 6 cm on a well-tracked GPS arc. Whatever a ray-traced grid file could add lives below that — and not all of the remainder is even radiation pressure. IGS finals are quoted near 2.5 cm, and the reference velocities used in the fit are differentiated from 15-minute sampled positions, which puts a floor under the residual unrelated to force modelling. Against that, rebuilding a generator means supplying validated spacecraft CAD geometry with per-surface materials and the ray-tracing physics over it — self-shadowing, multiple reflection, specular and diffuse separation. That content is absent, largely non-public, and unbounded in effort. Spending it to chase a few centimetres that may not be radiation-pressure-shaped is not a case anyone should fund.

One qualification, about who this applies to

An *estimated* parameter works when there is tracking data to fit against. A *physical* model earns its keep precisely when there is not:

- **Orbit prediction** — a parameter cannot be estimated from observations that have not been made yet. Predicted ephemerides, conjunction screening days ahead, re-entry.
- **Short or sparse arcs** — debris and newly detected objects, with a handful of observations rather than 87 epochs at 15-minute sampling.
- **High area-to-mass ratio objects**, where radiation pressure dominates the dynamics and the arcs are usually the worst available.

Well-tracked GNSS satellites were never the market for ray-traced radiation pressure — they are its *validation domain*, and there everyone estimates empirical parameters instead. If a market exists it is space situational awareness and debris, which is exactly what the Cone_HAMR, Large_Sphere_HAMR and Small_Sphere_LAMR property sets in the source were built for. “Not worth building” is a judgement about this evidence and this application, not a finding that physical radiation pressure modelling has no value.

Where the asset actually is now

It has moved, twice, and it is worth being explicit about where it landed.

Revision 1	The ray-traced SRP engine — scarce, and missing its fuel
Revision 2	The validation — a credential that makes the SRP claim falsifiable
Revision 3	A working, validated, parameter-estimating orbit determination capability. The grid files turned out to be substitutable by a parameter; the trustworthiness did not

The question a buyer asks first is not “can this model an exotic spacecraft”. It is “can this be trusted at all”. This tree can now answer the second question reproducibly, against public reference data, with a script.

What has been closed

- **Version control and provenance.** Under git, with 950 MB of atmosphere tables and IDE cache excluded and documented. PROVENANCE.md inventories six vendored components and ten datasets with their licences.
- **Two silent correctness defects.** The orbit fitter was building least squares on identity partials,

because only two of five propagators integrate the state transition matrix and the default is not one of them. And the Makefile tracked no header dependencies, so editing a header left other objects with the old struct layout — corrupting memory at run time rather than failing to compile. Both fixed; every number in this revision was taken afterwards.

- **Parameter estimation.** The seventh parameter, integrated rather than differenced, with the six-state path preserved and still reproducing its baseline exactly.

What still blocks a transaction

FOR THE USER not solvable in code

- **No licence or copyright statement anywhere in the first-party code**, which originates from UCL SGNL. Under standard UK university IP terms it belongs to UCL. Nothing can be sold, licensed or released until this is settled. **This is now the single largest obstacle** — larger than any remaining engineering gap.
- **CALCEPH is triple-licensed** — BSD-like, LGPL-like or GPL-like — a licensee must choose, no choice is recorded, and it is linked *statically* into every binary. It is optional and the build drops it cleanly, so shipping without it is a real alternative.

ENGINEERING what is genuinely still missing

- **No observation models.** It fits to already-determined precise orbits, not to raw tracking data. This is now the largest functional gap: it is what separates a force-model validation tool from an orbit determination system a customer could operate.
- **No covariance.** Rules out conjunction assessment and SSA — which, per the qualification above, is the one market where the physical modelling heritage would matter.
- **No manoeuvre modelling.** Single-threaded, one object per process. No Python bindings. IAU1980 frames rather than IERS 2010.

Valuation

- **The library as a sellable product: still approximately £0.** Orekit and GMAT are free, better governed and better tested. Nothing here changes that, and competing there remains the wrong move.
- **The validated capability: materially positive and newly real.** A reproducible 5.7 cm out-of-sample figure, with a physical property recovered independently, is a credential very few small teams can produce. It cost two days because the dynamics were already correct and had simply never been measured.
- **Services and know-how: still the only line item that could carry a business**, now with measured evidence rather than reputation. The credible offer is analysis, not software.
- **Replacement cost remains the wrong frame.** £300k–£600k to rebuild; you would adopt Orekit instead and port what is differentiated.

Recommendation

Do not archive it, do not productise it, and do not rebuild the ray-tracer. Finish proving the claim generalises, then decide on evidence.

1. **Broaden the validation table** — more GPS blocks, at least one Galileo arc, a LEO arc if a precise ephemeris can be obtained. Nearly free now, and it retires the one real risk in the current position: the 6 cm result rests on two satellites, one constellation, two consecutive days. *In progress.*
2. **Find the floor.** Quantify how much of the residual is method artefact rather than dynamics, then extend the estimation — not the modelling — to an ECOM-style set to see whether any

structured radiation-pressure residual survives at all. A finding that it does not would close the physical-modelling question permanently, which is worth knowing definitively. *Commissioned.*

3. **Settle the IP position in parallel.** It gates everything and needs no engineering.
4. **Then choose one of two ends.** If the table generalises and a customer exists, the offer is analysis services and the next engineering investment is observation models and covariance — the SSA direction, where the physical heritage still matters. If no customer exists, stop here: it is a well-documented, validated, publishable body of work with an accuracy figure attached, which is a respectable place to leave it.

What to find out first

Both remaining questions are yours rather than an engineer's, and both are slower than the engineering.

1. **Who owns this?** UCL's position on the 2014–2017 code, and the CALCEPH licence choice as a smaller instance of the same question.
2. **Is there a named customer?** Not for a propagator — for analysis. If yes, the validation table is the opening argument. If no, this is an archive with an unusually good accuracy figure, and that is a legitimate outcome rather than a failure.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026, including the Phase 1 and Phase 2 work carried out by the `odl_maintainer` session and verified here. The competitive and valuation sections are judgement, not repository evidence. Revision 3 supersedes revision 2, which still treated the ray-traced assets as the thing whose absence mattered most.